

Name: _____ Date: _____

Period: _____

Consider the rational function $r(x) = \frac{x^2 - x - 2}{x^2 - 4}$.

1. Factor the numerator and denominator completely.

Numerator: $x^2 - x - 2 = (x - 2)(x + 1)$

Denominator: $x^2 - 4 = (x - 2)(x + 2)$

2. List any shared zeros of the numerator and denominator.

Shared zero(s): $x = 2$

3. For each shared zero, compare multiplicities and classify as a **hole** or a **vertical asymptote**. Then state any additional VAs from non-shared zeros.

Zero	Num. mult. m	Denom. mult. n	Hole or VA?
$x = 2$	1	1	Hole ($m = n \geq 1$)
$x = -2$	0	1	VA (denom only)

4. Find the coordinates (c, L) of any holes. Show the simplification and substitution.

Simplified form: $\frac{x + 1}{x + 2}$

$L = \frac{2 + 1}{2 + 2} = \frac{3}{4}$ Hole at: $\left(2, \frac{3}{4}\right)$

5. Write the limit statement for the hole using proper notation.

$\lim_{x \rightarrow 2} r(x) = \frac{3}{4}$